How to Program *with* Shell Scripts: A Tutorial

Shell scripts are computer programs that are specially made to be run by the UNIX shell. Typically, shell scripts are used for manipulation, program execution and printing text. They enable the user to communicate with the OS, and vice versa.

Most of what we will cover in this article has a history in the UNIX/Linux world, right from the days of AT&T, Bell Labs and BSD (Berkeley Software Division), when variants of UNIX flavours were created and the shell started evolving. Today, we have many variants of shell and a lot of options to choose from.

Users need to understand their exact requirements before choosing the shell that suits the situation. Some of the well-known and widely used shells are:

- sh
- bash
- csh
- ksh
- pdksh
- tcsh

What is a shell?

A shell is a process which is, or can be, spawned. It is also an interpreter for computer OSs. Shells take the input from the user and pass it on to the OS and vice versa. All shell commands can be executed and used to carry out various executions and output generation, using scripting.

What is a spawn?

A spawn is a combination of *exec()* and *wait()* system calls/library functions, which uses external commands to run and provide the necessary output.

Note: A program in execution is called a process.

Note 2: The command *lsb_release –a* can be used to view the present version of the Linux variant. Throughout this document, I have used Ubuntu 17.04, also called 'zesty'. The command *uname –a* can be used to see the Linux kernel version, as well as more details.

```
ubuntu@ubuntu:~$ lsb_release -a
No LSB modules are available.
Distributor ID:    Ubuntu
Description:   Ubuntu 17.04
Release:   17.04
Codename:  zesty
ubuntu@ubuntu:~$

ubuntu@ubuntu:~$ uname -a

Linux ubuntu 4.10.0-19-generic #21-Ubuntu SMP Thu Apr 6
17:04:57 UTC 2017 x86_64 x86_64 x86_64 GNU/Linux
ubuntu@ubuntu:~$
```